

ZIYAO CHEN

EDUCATION

Northwestern Polytechnical University

M.Eng. in Computer Technology

Admitted with exemption from the National Entrance Examination

Advisor: Prof. Lingyun Song

Key Laboratory of Big Data Storage and Management, Ministry of Industry and Information Technology

GPA: 85.93/100

Xi'an, China

Sep. 2024 - Jun. 2027

First-Class Scholarship; Outstanding Master's Student Award

Hefei University of Technology

B.Eng. in Computer Science and Technology

School of Computer and Information

GPA: 85.76/100

Hefei, China

Sep. 2020 - Jun. 2024

First-Class Scholarship; University-Level Merit Student Award

RESEARCH INTERESTS

Continual Learning, Self-Evolving Agents, Multimodal Large Language Models, Knowledge Editing

PUBLICATIONS

- Lingyun Song, **Ziyao Chen**, et al. ROSE: Risk-Aware Orthogonal Subspace Navigation for Lifelong Knowledge Editing in Multimodal Large Language Models. **NeurIPS**, 2026. (**CCF-A; Spotlight; First student author; Spotlight rate: 0.95%**)
- Lingyun Song, **Ziyao Chen**, Kang Pan, et al. KSS-MoE: Knowledge Space Synergy Framework in Mixture of Experts for Continual Visual Instruction Tuning. **AAAI**, 2026. (**CCF-A; First student author; Acceptance rate: 17.6%**)
- Ziyao Chen**, Xuequn Shang, and Lingyun Song. Iteratively-Enhanced Dynamic Expansion Model for Continual Learning. **WISA**, 2024. (**CCF-C; Oral Paper**)

PATENTS

- Lingyun Song, **Ziyao Chen**, Xuequn Shang. An Iteratively-Enhanced Dynamic Expansion Continual Learning Model. CN120123765A, Feb. 2025 (Published).
- Ziyao Chen**, Hai Min, Yemao Zhang, Kai Shu, Han Chen. A Facial Landmark Detection Method Based on Mixed-Domain Attention Mechanism. CN115223228B, Aug. 2022 (Granted)
- Xuequn Shang, Binze Shi, Lingyun Song, **Ziyao Chen**, et al. Information Diffusion Prediction Method for Hybrid Graph Transformer Based on 2D Positional Encoding Enhancement. CN120541410A, May 2025 (Published)

RESEARCH EXPERIENCE

Research on Multi-Step Reasoning Methods for Complicated Visual Problems under Data Constraints

NSFC General Program (Graduate Researcher)

Jan. 2026 - Dec. 2029

- Led the continual adaptation direction for data-constrained multi-step visual reasoning, studying how multimodal models can acquire new knowledge without eroding prior reasoning capabilities.
- Designed risk-aware orthogonal knowledge-subspace updates that isolate new edits from previously learned subspaces, protecting stability-critical knowledge during long update sequences.
- Developed multi-granularity knowledge integration and semantic activation mechanisms to retrieve edited knowledge selectively while preserving model behavior on unrelated inputs.
- This work was accepted as a **Spotlight** at **NeurIPS 2026**.

Step-Level DPO for MoE LLMs in Continual Learning of Complex Multi-Domain Tasks

Xiaomi Open-Competition Research Program (Lead Graduate Researcher)

May 2025 - Jul. 2026

- Developed KSS-MoE for continual visual instruction tuning, using internal activations to fuse task-relevant low-rank subspaces from specialized experts for fine-grained collaboration at inference.
- Designed specialized and shared expert collaboration with conflict masks and orthogonal gradient projection to facilitate cross-task transfer while reducing negative interference.
- Evaluated the framework across all eight CoIN tasks, achieving state-of-the-art continual-learning performance; the work was published at **AAAI 2026**.

Multimodal Intelligent Continual Question Answering with Dynamic-KG RAG

CCF-Zhipu Large Model Innovation Fund (Graduate Researcher)

Nov. 2024 - Nov. 2025

- Investigated coordinated adaptation between evolving structured graph knowledge and multimodal model knowledge for temporal and cross-domain learning.
- Constructed visual, textual, multimodal, and shared expert subspaces to strengthen cross-domain knowledge transfer while retaining previously acquired knowledge.
- Introduced cross-subspace orthogonality constraints to decouple domain-specific and general knowledge optimization, improving generalization while mitigating forgetting and conflicts.

HONORS & AWARDS

- National First Prize, National Computer Competition for Colleges and Universities Dec. 2022
- Provincial First Prize, Chinese Collegiate Computing Competition (4C) (*Project Lead*) Aug. 2022
- Provincial First Prize, 16th iCAN Innovation Contest (*Project Lead*) Sep. 2022
- Provincial Silver Award, China International College Students' Innovation Competition (*Project Lead*) Jul. 2025
- Provincial Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM) Mar. 2023
- Provincial Undergraduate Innovation Program (*Project Lead*) May 2021 - Apr. 2022
- Recipient of **more than ten** national, provincial, and ministerial-level awards, including eight as project lead.